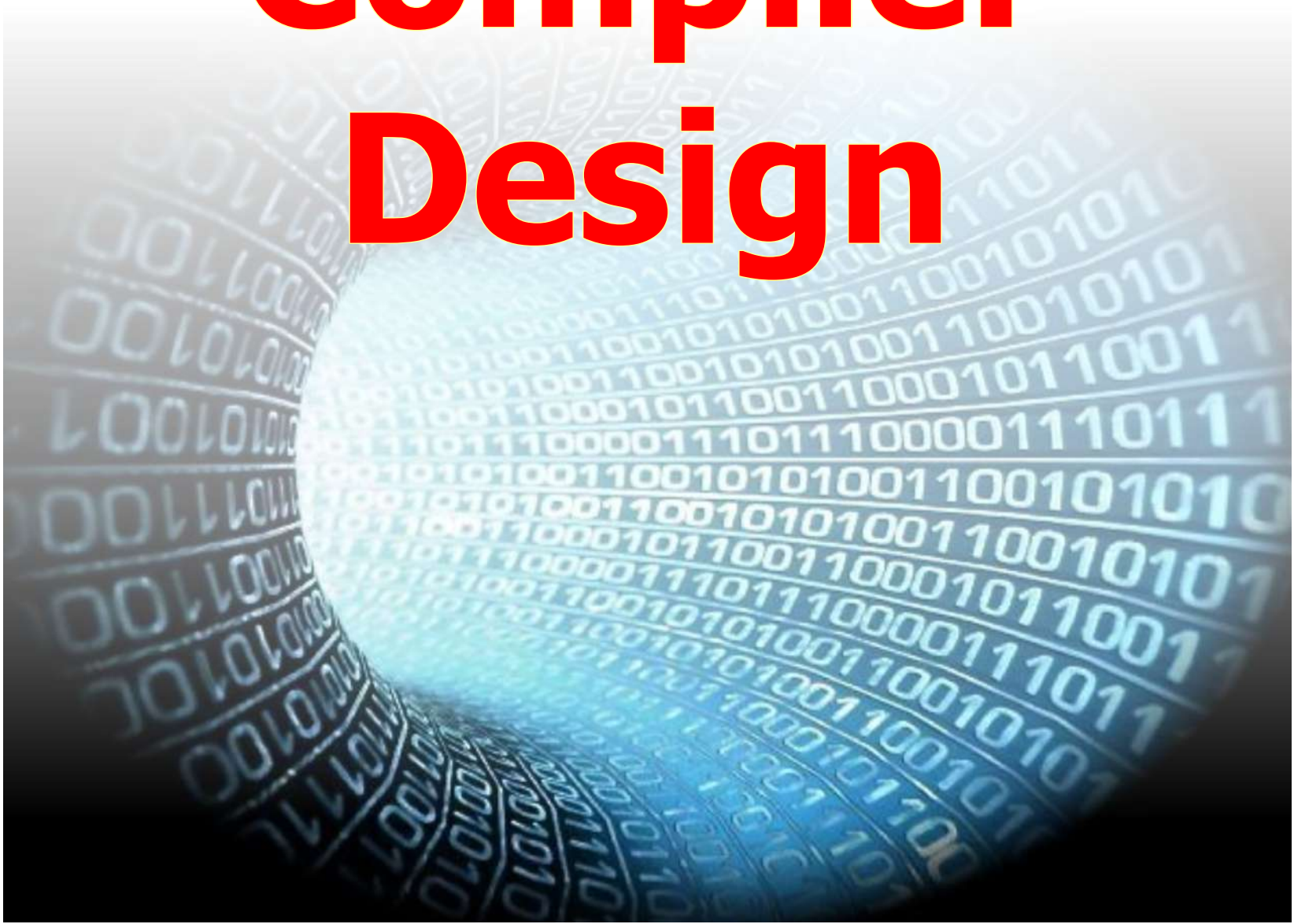




Compiler Design

The background features a perspective view of a tunnel formed by rows of binary code (0s and 1s). The code is rendered in a light blue color and recedes into the distance, creating a sense of depth. The overall design is clean and modern, with a teal header and a white-to-blue gradient background.

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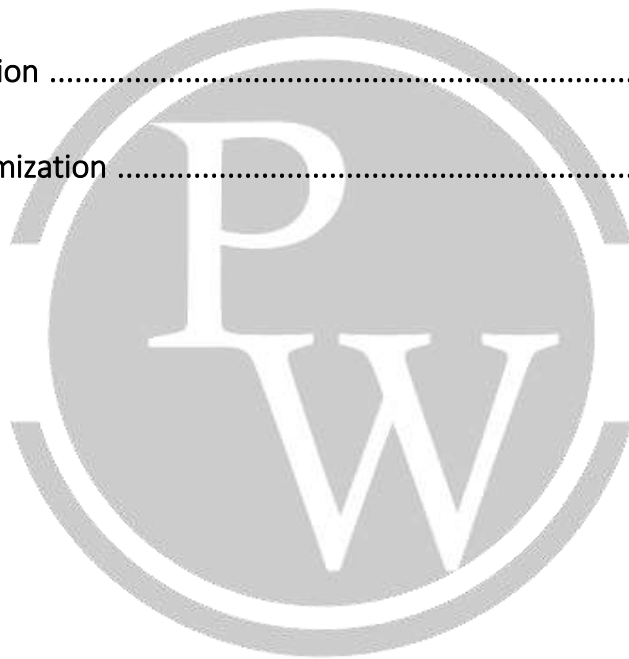
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Compiler Design

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1

INTRODUCTION AND LEXICAL ANALYSIS

1.1 CD Introduction

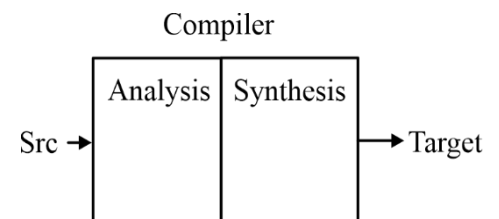
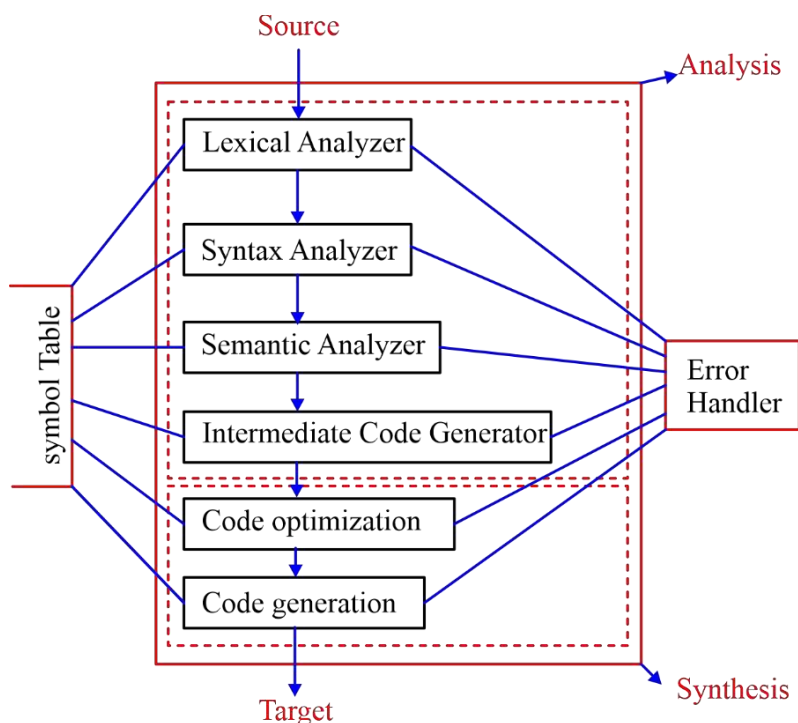
1.1.1 Definition

Convert High Level Language to Low Level Language.



- High Level Language can perform more than one operation in a single Statement.
- Low Level Language can perform at most one operation in a statement.

1.2 Analysis and Synthesis model of Compiler



- There are 6 phases of the Compiler



1. Lexical Analyzer:

- Program of DFA, it checks for spelling mistakes of program.
- Divides source code into stream of tokens.

2. Syntax Analyzer:

- Checks grammatical errors of the program (Parser).
- Parser is a DPDA.

3. Semantic Analyzer:

Checks for meaning of the program.

Example:

Type miss match, stack overflow

4. Intermediate Code Generation:

- This phase makes the work of next 2 phases much easier.
- Enforces reusability and portability.

5. Code optimization:

- Loop invariant construct
- Common sub expression elimination
- Strength Reduction
- Function in lining

Deadlock elimination

6. Symbol Table:

- (1) Data about Data (Meta data)
- (2) Data structure used by compiler and shared by all the phrase.



2

PARSING

2.1 CD - Grammar

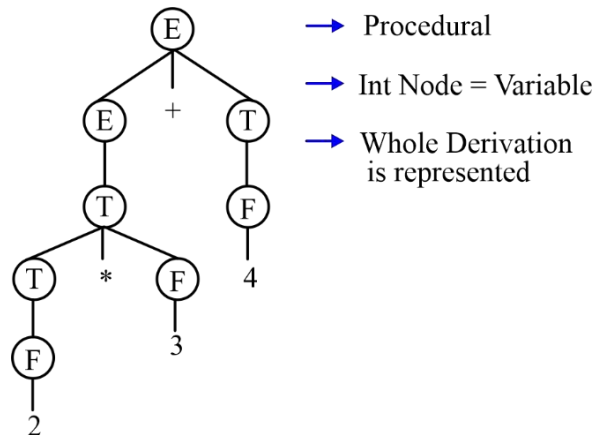
- In compiler we only use: Type – 2 (CFG) and Type – 3 Regular grammar.
- Compiler = Program of Grammar
- Compiler = Membership algorithm
- Every programming Language is Context Sensitive Grammar (Context Sensitive Language)

2.1.1 Parse Tree and Syntax Tree

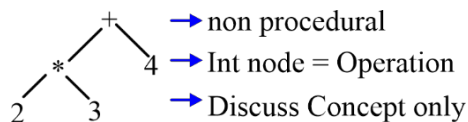
G: $E \rightarrow E + T / T$ $E \rightarrow E + T \rightarrow T + T \rightarrow T * F + T \rightarrow F * F + T \rightarrow 2 * F + T \rightarrow 2 * 3 + T \rightarrow 2 * 3 + F$
 $T \rightarrow T * F / F$

$E \rightarrow 2 * 3 + 4$

Parse Tree:

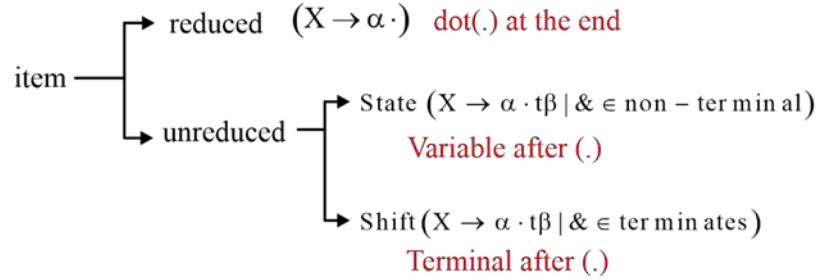


Syntax Tree:



- To check to priority / Associativity:
Randomly derive till you have enough operators, then check which one is done first.
- If priority of 2 operators is same and both Left and Right associative → **Ambiguous Grammar** [Useless]

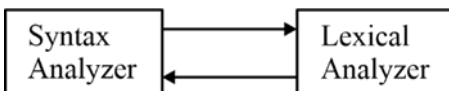
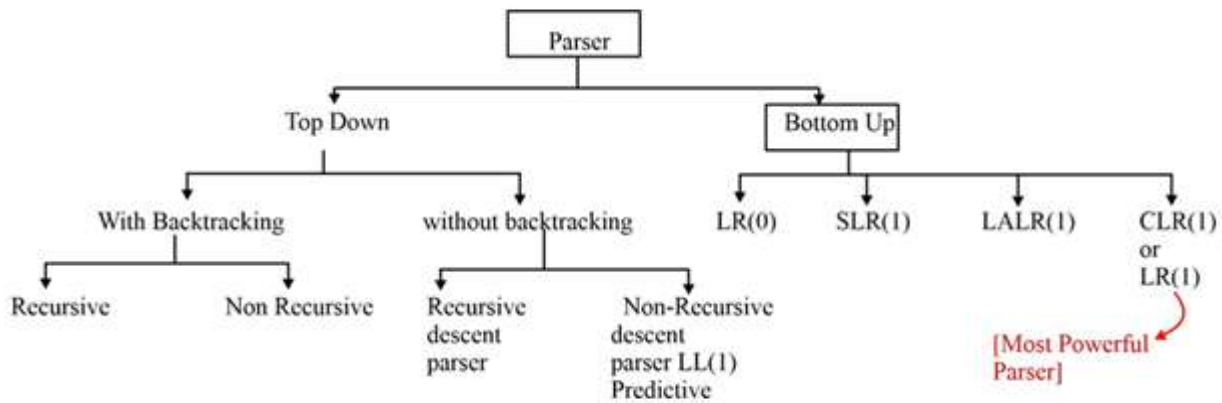
2.1.2 Types of Parsers in Bottom Up: [CD - Parser]



2.2 CD-Syntax Analysis / Parsing

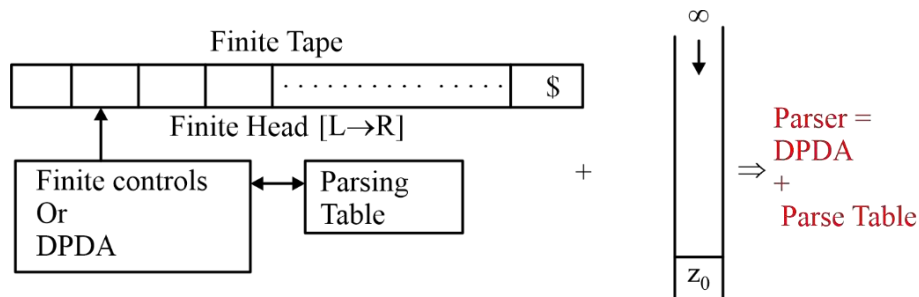
Grammatical Errors are checked by the help of parsers.

- Parsers are basically DPDA



- All of these parsers are table driven.

2.2.1 Mathematical Model of Parser



- Parsers generate Parse Tree, for a given string by the given grammar.

2.2.2 Top-Down Parser (LL (1))

- It uses LMD and is equivalent to DFS in graph.

2.2.3 Algorithm to construct Parsing Table

1. Remove Left Recursion if any.
 2. Left Factor [Remove Common Prefix]
 3. Find first and follow set
 4. Construct the table.
- If we increase the look ahead symbol:

2.2.4 Removal of common Prefix: (Left Factor)

1. $S \rightarrow a \mid a b \mid a A$ $S \rightarrow a Y$
 $Y \rightarrow \epsilon \mid b \mid A$
2. $A \rightarrow a b A \mid a A \mid b$
 $A \rightarrow a \times \mid b$ $A \rightarrow a \times \mid b$
 $X \rightarrow b A \mid A$ $X \rightarrow b A \mid a x \mid b$
 $A \rightarrow a X \mid b$
 $X \rightarrow a X \mid b$
 $Y Y \rightarrow A \mid \epsilon$

2.2.5 First and Follow

- First Set $\rightarrow$ extreme Left terminal from which the string of that variable starts.
 $\rightarrow$ it never contains variable, but may contain ' ϵ '.
 $\rightarrow$ we can always find the first of any variable.
- Follow set $\rightarrow$ Follow set contains terminals and \$.
It can never contain variable and ' ϵ '.
How to find follow set?
 1. Include \$ in follow of start variable.
 2. If production is of type $\rightarrow$
 $A \rightarrow \alpha B \beta$ $\alpha \beta \rightarrow$ strings of grammar symbol].
 $\text{follow}(B) = \text{first}(\beta)$.
If, $\beta \rightarrow \epsilon$, i.e. $A \rightarrow \alpha B$, then $\text{follow}(B) = \text{follow}(A)$.
- Productions like: $A \rightarrow \alpha A$ gives no follow set.

2.2.6 Example of first and follow set

- $S \rightarrow AB \mid CD$
 $A \rightarrow aA \mid a$
 $B \rightarrow bB \mid b$
 $C \rightarrow cC \mid c$
 $D \rightarrow dD \mid d$

	First	Follow
S	a, c	\$
A	a	b
B	b	\$
C	c	D
D	D	\$

2.2.7 Entity into Table: Top Down

- No of Rows = number of unique variable in Grammar.
 - No of columns = [Terminals + \$]
 - For a Variable (Row) fill the column (Terminal) if its P, there in its first with the production required.
 - If ϵ is in first put $V \rightarrow \epsilon$ under \$ and its follow.
- If any cell has multiple items, then its not possible to have LL (1) Parser, since that will be ambiguous.
 - In top down we do: Derivation
In Bottom up we do: Production.

Question: Construct LL (1) Parsing Table for the given Grammar: $E \rightarrow E + T \mid T$; $T \rightarrow T * F \mid F$; $F \rightarrow (E) \mid id$; $\Rightarrow G_0$

$\Rightarrow$ Removing Left Recursion:

$E \rightarrow TE'$

$E' \rightarrow + TE' \mid \epsilon$

$T \rightarrow FT'$ G_1

$T' \rightarrow * FT' \mid \epsilon$

$F \rightarrow (E) \mid id$

	First	Follow
E	C, id	\$,)
E'	+, ϵ	\$,)
T	C, id	+, \$,)
T'	*, ϵ	+, \$,)
F	C, id	*, +, \$,)

- Left Factoring not Required:
Construction of Table: [LL (1)]

	+	*	(	)	id	\$
E	Error	Error	$E \rightarrow TE'$	Error	$E \rightarrow TE'$	Error
E'	$E' \rightarrow TE'$	Error	Error	$E' \rightarrow \epsilon$	Error	$E' \rightarrow \epsilon$
T	Error	Error	$T \rightarrow FT'$	Error	$T \rightarrow FT'$	error
T'	$T' \rightarrow \epsilon$	$T' \rightarrow * FT'$	Error	$T' \rightarrow \epsilon$	Error	$T' \rightarrow \epsilon$
F	Error	Error	$F \rightarrow (E)$	Error	$F \rightarrow id$	error

- Since for G_1 , Table constructed with no multiple entries. Hence successfully completed. Hence G_1 is LL (1)

Question: Construct LL (1) Parsing Table for the following Grammar:

$S \rightarrow L = R \mid R$; $L \rightarrow * R \mid id$; $R \rightarrow L \Rightarrow G_0$

Solution: Left Factoring:

$$\begin{aligned}
 S &\rightarrow L = R \mid L & S &\rightarrow LX \\
 L &\rightarrow *R \mid id & \Rightarrow & \times \Rightarrow = R \mid \in \\
 R &\rightarrow L & L &\rightarrow *R \mid id \\
 & & R &\rightarrow L
 \end{aligned}
 \Rightarrow G_1$$

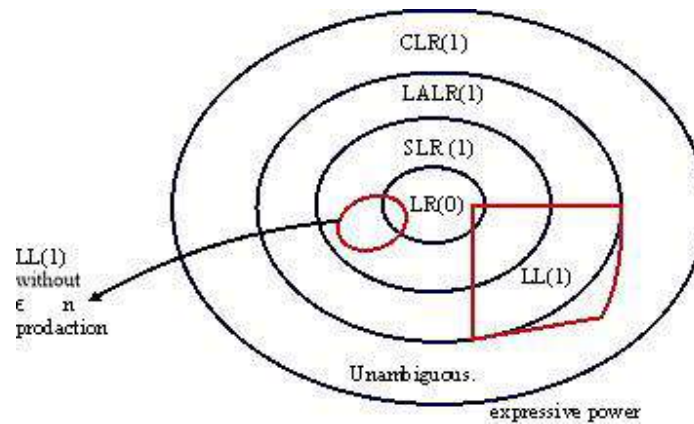
	First	Follow
S	*, id	\$
×	=, ∈	\$
L	*, id	\$
R	*, id	\$

Construction of Table:

	*	=	Id	\$
S	S→LX	Error	S→L X	Error
L	L→*R	Error	L→id	Error
R	R→L	Error	R→L	Error
×	Error	X→ = R	Error	X→∈

- G_1 is a LL (1) Grammar.

2.2.8 Hierarchy of Parsers: [for ϵ - free Grammar]



- For ϵ -producing grammars every LL (1) may not be LALR (1).

Note:

We can't construct any parser for ambiguous grammar. Except: Operator precedence, parser possible for some ambiguous grammar.

Example:

$G: S \rightarrow a S a \mid b S b \mid a \mid b$ (Unambiguous but no parser) $L(G) = \omega(a + b) \omega R$
(Odd palindrome)

- Every RG is not LL (1) as it may be ambiguous, or Recursive or Common Prefix.
- Parsers exists only for the grammar if its Language is DCFL.
- There are some grammars whose Language is DCFL but no parser is possible for it.

2.4.9 Operator Precedence Grammar

Format: (1) No 2 or more variable side by side
(2) No $\in$ production.

Example:

$$\begin{array}{lll} E \rightarrow E + T \mid T & E \rightarrow E + E \\ T \rightarrow T * F \mid F & E \rightarrow E * E & S \rightarrow aSa \mid bSb \mid a \mid b \\ F \rightarrow (E) \mid id & \text{or } E \rightarrow a / b & \text{or O.G} \\ \text{O.G.} & \text{O.G} & \\ & \text{Or} & \\ S \rightarrow AB & & \\ A \rightarrow aA \mid \epsilon & \text{Not O.G} & \\ B \rightarrow bB \mid \epsilon & & \end{array}$$

2.2.10 Checking LL (1) without Table

$$\begin{array}{ll} A \rightarrow \alpha_1 \alpha_2 \alpha_3 \text{ then } \rightarrow & A \rightarrow \alpha_1 \alpha_2 \alpha_3 \mid \epsilon \\ \text{first}(\alpha_1) \cap \text{first}(\alpha_2) = \emptyset & \text{first}(\alpha_1) \cap \text{first}(\alpha_2) = \emptyset \\ \text{first}(\alpha_1) \cap \text{first}(\alpha_3) = \emptyset & \text{first}(\alpha_1) \cap \text{first}(\alpha_3) = \emptyset \\ \text{first}(\alpha_2) \cap \text{first}(\alpha_3) = \emptyset & \text{first}(\alpha_2) \cap \text{first}(\alpha_3) = \emptyset \\ & \text{follow}(A) \cap \text{first}(\alpha_1) = \emptyset \\ & \text{follow}(A) \cap \text{first}(\alpha_2) = \emptyset \\ & \text{follow}(A) \cap \text{first}(\alpha_3) = \emptyset \end{array}$$

2.2.11 Bottom-UP Parser

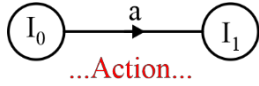
- It uses RMD in reverse and has no problem with:
 - (a) Left recursion (b) Common Prefix
- SLR (1) LR (0) CLR (1) LR (1)
- LR (0) items LALR (1) items
- LR (1) = LR(0) + 1 Lookahead
- No parser possible for ambiguous grammar.
- There are some unambiguous grammars for which, there are no Parser.

2.2.12 Basic Algorithm for Construction

- Augment the grammar and expand it, and give numbers to it
- Construct LR (0) or LR (1) item set.
- From that fill the entries in the Table Accordingly.

2.2.13 Types of Entries

(1) **Shift Entries:** Transitions or Terminals



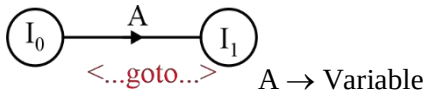
Entry:

	a
I ₀	S ₁

$a \in \text{Terminal } I_0, I_1: \text{Item sets.}$

- Shift entries are same for all Bottom – up Parser.

(2) **State Entry:** Transition or non – terminal (variable)



Entry:

	A
I ₀	1

- Same for all Bottom-up Parser.

(3) **Reduce Entity:** done for each separate production in the item set of type:

$$i > \times \rightarrow \alpha$$

where, $i \rightarrow \text{Production Number}$

$\times \rightarrow \text{Producing Variable}$

$\alpha \rightarrow \text{Grammar String}$

(a) LR (0) Parser:

Put Ri in every cell of the set-in action table (**ALL**).

(b) SLR (1) Parser:

Put Ri only in the follow(x) form the Grammar. (**Follow(x)**)

(c) LALR (1) and CLR (1):

Put Ri only in the look ahead of the production (**Lookaheads**)

2.2.14 Conflicts

LR (0) Parser:

SR: Shift Reduce Conflict

$S \rightarrow X \rightarrow \alpha \cdot t\beta \Rightarrow$ then SR

$R \rightarrow Y \rightarrow \delta \cdot$ conflict

RR: Reduce Reduce conflict

$R \rightarrow X \rightarrow \alpha \cdot \Rightarrow$ then RR

$R \rightarrow X \rightarrow \beta \cdot$

SLR (1) Parser:

SR:

$S \rightarrow X \rightarrow \alpha \cdot t\beta \quad \$ \Rightarrow$ then SR

$R \rightarrow Y \rightarrow \delta \cdot$ conflict

$t \in \text{follow}(Y)$

RR:

$X \rightarrow \alpha \cdot \quad \$ \Rightarrow$ then RR

$Y \rightarrow \beta \cdot$

$\text{Follow}(x) \cap \text{follow}(y) \neq \emptyset$

LALR (1) and CLR (1): Same as SLR (1), but instead

SR:

$X \rightarrow \alpha \cdot t\beta \cdot L_1 \quad \$ \Rightarrow$ then SR

$Y \rightarrow \delta \cdot L_2$ conflict

$T \in L_2$

RR:

$X \rightarrow \alpha \cdot, L_1 \quad \$$ then RR

$Y \rightarrow \beta \cdot, L_2$ conflict

$L_1 \cap L_2 \neq \emptyset$

2.4.15 Inadequate Static

A static having ANY conflict is called a conflicting state or independent state.

Note:

The state $S' \rightarrow S$ or $S' \rightarrow S, \$$ is accepted state, and this is not a reduction.

- The only difference between CLR (1) and LALR (1) is that, the states with the similar items, but different Lookaheads are merged together to Reduce space.

Important Points

- If CLR (1) doesn't have any conflict, then conflict may or may not arise after merging in LALR (1).
- If LALR (1) has SR – conflict, then we can conclude that CLR (1) also has SR – Conflict.
- LALR (1) has SR – conflict if and only if CLR (1) also has SR.



- We can construct parser for every unambiguous regular grammar: [CLR (1) Parser].

	L Left to right scan of i/p	R Using reverse RMD	(0) No Lookahead	
S	L	R	(1)	
Simple	L to R Scan	Using Reverse RMD	Lookahead	
L	A	L	R	(1)
Look	Ahead	L to R Scan	Revers RMD	Lookahead
C	L	R	(1)	
Canonical	L to R Scan	reverse RMD	Lookahead	

Very Important Point:

LALR (1) Parser can Parse non LALR (1) grammar, when only non-SR – Conflict by favouring shift over reduce.

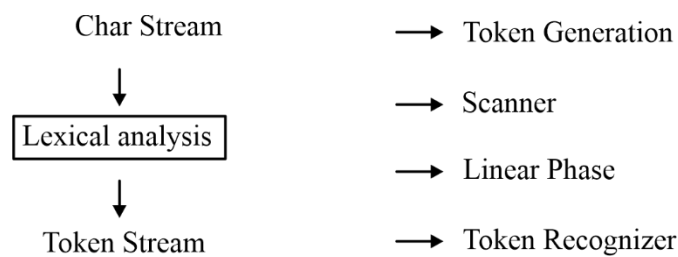
Example: $E \rightarrow E + E \mid E * E \mid id \mid 2 + 3 * 5 \Rightarrow E + E \cdot * 5$



3

LEXICAL ANALYSIS

3.1 Lexical Analysis



3.1.1 Definition

Scan the whole program, char by char and produces the corresponding token.

- Also produces /Lexical Errors (if any).
- Functions of Lexical Analyzer.
 - (1) Scans all the character of the program.
 - (2) Token recognizer.
 - (3) Ignores the comment & spaces.
 - (4) Maximal Munch Rule [Longest Prefix Match].

Note:

The Lexical Analyser uses, the Regular Expression.
Prioritization of Rules.
Longest Prefix match .

- Lexeme → smallest unit of program or Logic.
- Token → internal representation of Lexeme.

3.1.2 Types of Token

- (1) Identifier
- (2) Keywords
- (3) Operators
- (4) Literals/Constants
- (5) Special Symbol



3.1.3 Token Separation

- (1) Spaces
- (2) Punctuation

3.1.4 Implementation

- LEX tool $\Rightarrow$ Lex.yy.e
- All identifiers will have entry in symbol Table/ LA, gives entries into the symbol Table.
Regular Expression $\rightarrow$ DFA $\rightarrow$ Lexical Analyzer

3.1.5 Find number of Tokens

- (1)

```
void main () {  
    Printf("gate");  
}
```

{ 11 Tokens}
- (2)

```
int x, *P;  
    x = 10; p = &x; x++;
```

[18 Tokens]
- (3)

```
int x;  
    x = y;  
    x == y;
```

[11 Tokens]
- (4)

```
int 1x23;
```

[Lexical Error]
- (5)

```
char ch = 'A';
```

[5 Token]
- (6)

```
char ch = 'A;
```

Lexical Error.
- (7)

```
char *p = "gate";
```

[6 Tokens]
- (8)

```
char * p = "gate;
```

Error.
- (9)

```
int x = 10;  
/* comment x = x + 1; ERROR  
x = x + 1;
```

[11 Tokens]



4

SYNTAX DIRECTED TRANSACTION

4.1 Syntax Directed Transaction

CFG + Transition }
Syntax + Transition } SDT

SDT: CFG + Transition → 1) Meaning 2) Semantic

4.1.1 Application of SDTL

- (1) Used to perform Semantic Analysis.
- (2) Produce Parse Tree.
- (3) Produce intermediate representation.
- (4) Evaluate an expression.
- (5) Convert infix to prefix or postfix.

Example: $S \rightarrow S_1 S_2$ [$S.count = S_1.count + S_2.count$]

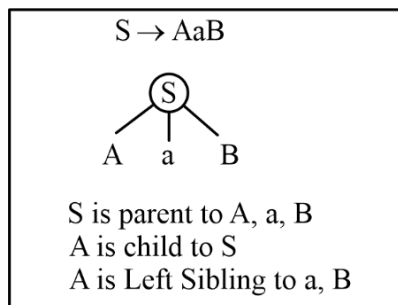
$S \rightarrow (S_1)$ [$S.count = S_1.count + 1$]

$S \rightarrow \epsilon$ [$S.count = 0$]

- Count is an attribute for non-terminal

4.1.2 Attributes

- (1) Inherited Attribute
- (2) Synthesized Attribute



- (1) Inherited Attribute: (RHS)

$S \rightarrow \boxed{A} B$ { $A.x = f(B.x \mid S.x)$ }

- The computation at any node (non-terminal) depends on parent or siblings (S).

- In above Example, x is inherited attribute.

(2) Synthesized Attribute: (LHS)

$$S \rightarrow AB \{S.x = f(A.x \mid B.x)\}$$

x is synthesized attribute.

The computation at any node (non-terminal) depends on children.

4.1.3 Identifying Attribute Type

- Always check every Translation.

- | | | |
|---|---|--|
| 1) $D \rightarrow T : L ; \{L.Type = T.Type\}$ inherited
$L \rightarrow L1, id ; \{L1.Type = L.Type\}$ inherited
$L \rightarrow id$
$T \rightarrow integer \{T.Type = int\}$ synthesized | } | Type is neither inherited nor synthesized |
| 2) $E \rightarrow E1 + T \{E.val = E1.val + T.val\}$ synthesized
$E \rightarrow T \{E.val = T.val\}$ Synthesized
$T \rightarrow T1 - F \{T.val = T1.val - F.val\}$ Synthesized
$F \rightarrow id \{F.val = id.val\}$ synthesized | } | Val is synthesized attribute |
| 3) $S \rightarrow AB \{Aa = B.x; S.y = A.x\}$ x is inherited y is synthesized
$A \rightarrow a \{A.y = a\}$ y is synthesized
$B \rightarrow b \{B.y = a.y\}$ y is synthesized | } | $x \Rightarrow$ inherited
$y \Rightarrow$ synthesized |
| 4) $D \rightarrow T L \{L.in = T.type\}$ inherited(in)
$T \rightarrow int \{T.type = int\}$ /synthesized
$L \rightarrow id \{Add\ type(id.entry, L.in)\}$ | } | $in \Rightarrow$ inherited
$type \Rightarrow$ synthesized |

4.1.4 Syntax Directed Definitions (SDDs): (Attribute Grammar)

(1) L-Attributed Grammar

- Attribute is synthesized or restricted inherited. (1) Parent 2) Left sibling only).
- Translation can be appended anywhere is RHS of production.
- Example: $S \rightarrow AB \{A.x = S.x + 2\}$
 or, $S \rightarrow AB \{B.x = f(A.x \mid S.x)\}$
 or, $S \rightarrow AB \{S.x = f(A.x \mid B.x)\}$
- Evaluation: In Order (Topological).

(2) S-Attributed Grammar:

- Attribute is synthesized only.
- The transaction is placed only at the end of production.
- Example: $S \rightarrow AB \{S.x = f(Ax \mid Bx)\}$.
- **Evaluation:** Reverse RMD (Bottom-Up Parsing).

4.1.5 Identify SDD

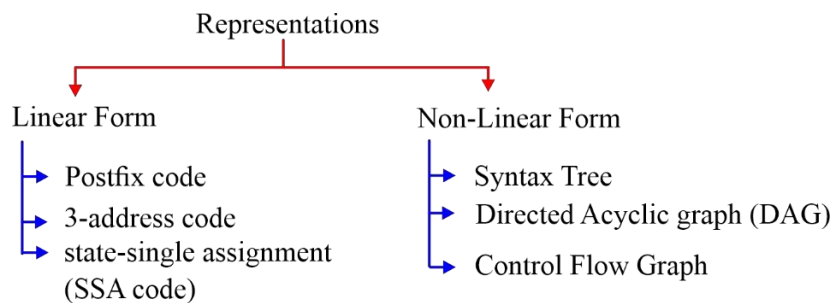
- (1) $E \rightarrow E_1 + E_2$ {E.type = if (E₁.type == int & & E.type == int) then int} Synthesizer else type – error.
 $E \rightarrow id$ {E.type = Lookup (id.entry)} synthesizer.
- type is synthesized, hence S-attribute and also L-attributed Grammar.
 - Every S-attributed Grammar is also L-attributed Grammar.
 - For L-attributed Evaluation, use the In-order of annotated Parser Tree.
 - For S-attributed, reverse of RMD is used.
- find RMD Order.
 → Consider its reverse.



5

INTERMEDIATE, CODE OPTIMIZATION

5.1 Introduction



Example Expression:

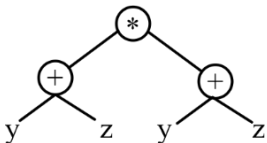
$(y + z) * (y + z)$

Post fix $\rightarrow$ $yz + yz + *$

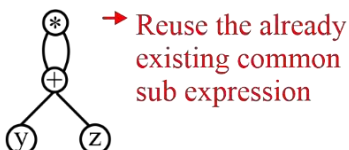
SSA $\rightarrow$ $t_1 = y + z$
 $t_2 = t_1 * t_2$ $\Rightarrow f_1$ and f_2 cannot be reassigned

3AC $\rightarrow$ $t_1 = y + z$
 $t_2 = t_1 \times t_2$

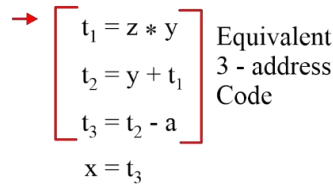
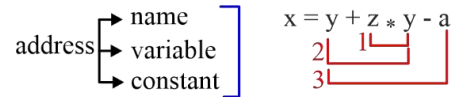
Syntax Tree $\rightarrow$



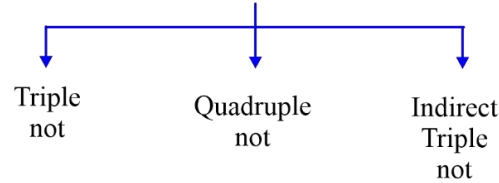
DAG $\rightarrow$



3-address Code: Code in which, at most 3 addresses. [including LHS]



Representation of 3AC



0	*	z	y		0	*	z	y	t_1	6	(0)	
1	+	y	(0)		1	+	y	t_1	t_2	7	(1)	
2	-	(1)	a		2	-	t_2	a	t_3	8	(2)	
3	=	(2)			3	=	t_3		x	9	(3)	

Indirect notation pointers to the numbers of Triple

Triple Notation → Space efficient → time inefficient

Quadruple → space inefficient → time efficient

- 3AC done using operator precedence.

Find minimum number of variables required in equivalent 3AC:

1.

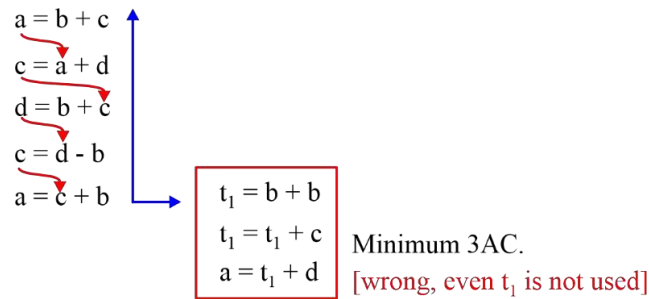
$$\begin{aligned}
 & x = u - t \\
 & y = x * v \\
 & x = y + w \\
 & y = t - z \\
 & y = *x * y \\
 & \text{7 variables}
 \end{aligned}$$

$$\begin{aligned}
 x * y &\Rightarrow x * (t - z) \Rightarrow (y + w) * (t - z) \\
 &\Rightarrow ((x * v) + w) * (t - z) \\
 &\Rightarrow (((u - t) * v) + w) * (t - z)
 \end{aligned}$$

$$\left[\begin{array}{ll} u = u - t & u = t - z \\ v = u * v & z = w * v \\ w = v + w & \end{array} \right] \Rightarrow 5 \text{ variables only}$$

⇒ Evaluating the expression:

$$\begin{aligned}
 a &\Rightarrow c + b \Rightarrow d - b + b \Rightarrow b + c - b + b \\
 &\Rightarrow b + a + d - b + b \\
 &\Rightarrow b + b + c + d - b + b \\
 &\Rightarrow b + b + c + d
 \end{aligned}$$



∴ Minimum:

$$\left\{ \begin{array}{l} b = b + b \\ c = b + c \\ a = c + d \end{array} \right\} \Rightarrow \text{only 3 variables [most optimal]}$$

5.1.1 Static Single Assignment Code: (SSA Code)

Every variable (address) in the code has single assignment [single meaning] + 3AC.

1. $x = u - t$
 $y = x * u$
 $x = y + w$
 $y = t - z$
 $y = x * y$

Find SSA?

⇒ $[u, t, v, w, z]$ are already assigned so we can't use them.

Equivalent SSA Code:

$x = u - t$ $y = x * u$

$p = y + w$

$q = t - x$

$r = p * q$

in use: $x, y, p, q, r \Rightarrow$ additional

∴ Total Variable ⇒ 10.

2. $p = a - b$

$q = p * c$

$p = u * v$

$q = p + q$

⇒ $[a, b, c, u, v]$ are already assigned,

Equivalent SSA Code:

$p = a - b$

$q = p * c$

$p_1 = v * u$

$q_2 = p_1 + q$

in use: p, q, p_1, q_2

∴ Total Variable = 9

5.1.2 Control Flow Graphs

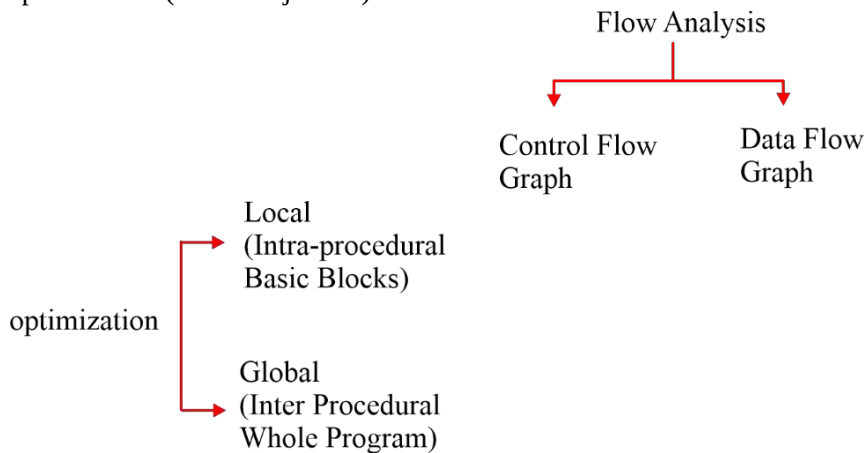
- CFG contain group of basic blocks and controls. CFG has nodes and edges to define basic blocks and controls.
- Basic Blocks: Sequence of 3-address code statements, in which control enters only from 1st statement (called as leader), and leaves from last statement.

Example:

1. $i = 1$	}	LB ₁
2. $j = 1$		LB ₂
3. $t1 = 5 * i$	}	LB ₃
4. $t2 = t1 + 5$		
5. $t3 = 4 * t2$		
6. $t4 = t3$		
7. $a[t4] = 1$	}	LB ₄
8. $j = j + 1$		
9. $\text{if } j \leq 5 \text{ goto } 3$	}	LB ₅
10. $i = i + 1$		
11. $\text{if } i < 5 \text{ goto } 2$	}	LB ₆

5.2 Code Optimization

Saves space / time. (Basic Objective)



5.2.1 Optimization Methods

- Constant folding
- Copy propagation
- Strength Reduction
- Dead code elimination
- Common sub expression elimination.
- Loop Optimization
- Peephole Optimization

1. Constant Folding

(i) $x = 2 * 3 + y \Rightarrow x = 6 + y$

Folding

ii) $x = 2 + y * 3$ can't fold the constant

2. Copy Propagation

i) Variable Propagation:

$x = y;$

$z = y + 2;$

$\Rightarrow z = x + 2;$

ii) Constant Propagation:

$x = 3$

$z = 3 + a;$

$\Rightarrow z = x + a$

3. Strength Reduction:

Replace expensive statement / instruction with cheaper one.

(i) $x = 2 * y$ **costly** $\Rightarrow x = y + y$; **Cheap**

(ii) $x = 2 * y \Rightarrow x = y < 1$; **Much Cheaper**

(iii) $x = y / 8 \Rightarrow x = y > > 3;$

4. Dead Code Elimination:

$x = 2$; **FALSE**

if ($x > 2$)

printf("code"); **DEAD CODE CAN BE REMOVED**

else

printf("optimization");

$x = 2;$
printf("optimization");

- Hence, above dead code never executes during execution. We can always delete such code.

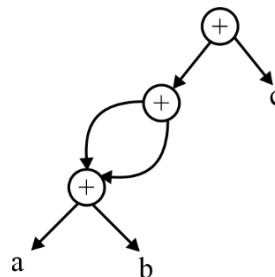
5. Common Sub Expression Elimination:

DAG is used to eliminate common sub expression.

Example: $x = (a + b) + (a + b) + c;$

$\Rightarrow t_1 = a + b$

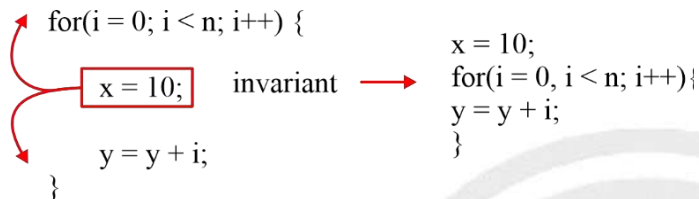
$x = t_1 + t_1 + c$



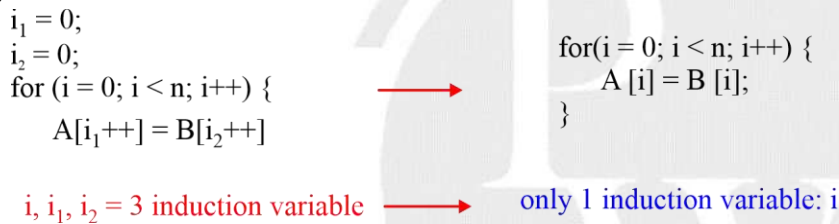
6. Loop Optimization:

(i) Code Motion – Frequency Reduction:

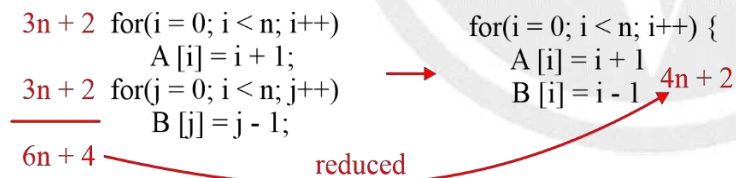
Move the loop invariant code outside of loop.



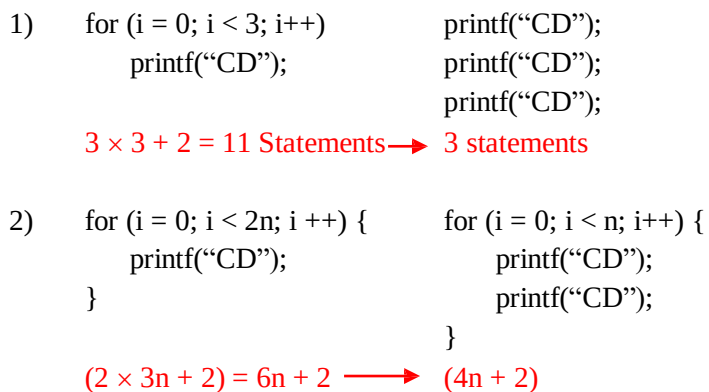
(ii) Induction Variable elimination:



(iii) Loop Merging / Combining: (Loop Jamming)



(iv) Loop Unrolling:



7. Peephole Optimization:

Examines a short sequence of target instructions in a window (*peephole*) and replaces the instructions by a faster and/or shorter sequence when possible.

- Applied to intermediate code or target code
- Following Optimizations can be used:



- Redundant instruction elimination
- Flow-of-control optimizations
- Algebraic simplifications
- Use of machine idioms

